

Information on the Programming Project

Machine Learning PS (SS 2026)

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Read the following information carefully!

I. OVERVIEW OF THE PROJECT

A. *First Things First*

Start by reading this description thoroughly. Then, before you start working on the dataset, thoroughly read through the report template and start discussing what the questions in the template require you to do.

It goes without saying that the rules of academic honesty apply. In short, this means doing your own work and not cheating, not presenting the work of others as your own and not submitting previously submitted work. The use of Large Language Models is allowed to help with programming, but you must understand your code. You will be required to briefly present and explain it to your proseminar instructor and answer some basic comprehension questions about it in the last proseminar session.

You are not allowed to collaborate on the report outside your designated groups.

B. *Problem Selection and Assignment*

In this project your task is to solve a problem using machine learning methods. There are three different problems with respective datasets. The datasets will be assigned so that all the problems are covered evenly. You can and should state your preference and we will try to accommodate that. The datasets are available in the Project folder on **OLAT**. The notebook stubs show how to load the datasets and at the end these are also used to test your methods/your solution. Some basic information about the datasets is also provided in the notebooks.

C. *Report and Implementation*

To solve the machine learning problem, *principled* experimentation is necessary, i.e., you will have to try different methods and evaluate which work best. This experimentation and your results, must be described in the report. Use the provided **report template**. The report template contains questions. Note that we will grade your report based on how you answer those questions. The report must be two pages (three pages for the three-person team) in length; see the report-template for details.

An example project report is provided to serve as a guide for your project. In addition to following the report required, it also contains a model card for one of the methods implemented. This based on the paper “Model Cards for Model Reporting” by Mitchell *et al.* and is included to give an example of standardized model reporting in more

applied scenarios. It is highly encouraged to read this paper for insight and ideas on how to evaluate and transparently report on your machine learning model. However, you are not required to submit a model card for this project as your report should cover most of the information relevant for this course.

D. *Comparing Methods*

First you will need to investigate the dataset to decide what the problem is about and what methods can and should be applied.

Next, you will need to try *different* methods. Two methods are *different* if they are related to different PS topics, i.e., logistic regression vs. neural networks vs. Gaussian mixture models are three different classification methods. Neural networks with different activation function are *not* different methods.

You have to experiment with different methods (at least two methods; for teams with three people: at least three different methods). You can re-use anything from the PS notebooks, and any methods that are **already installed and available on our JupyterHub** (e.g., NumPy, scikit-learn, PyTorch).

Note that you will have to understand the method you use, be able to explain how it works, and most importantly explain what you used. Somebody else has to be able to replicate your results from the description – without knowing the content of the PS (but you can assume knowledge of the VO).

Obviously the description is easier for methods that we have covered. If the method is different from what we covered in the VO, you need to explain how it is different, i.e. how we can get from the methods we already know to the method you used.

Your goal is to solve the machine learning problem as adequately as possible. As opposed to the notebook assignments, in this assignment you can use libraries like scikit-learn, allowing you to focus more on finding a good model and following a principled way of achieving this (e.g., VO’s slide set on “Neural Networks C.3 – Generalization and Validation”).

For each method, different settings are possible, for example different kernels, different regularizers. Finding the right settings in a principled way is essential for machine learning.

3-Person Team: Since there are more people in this team, there needs to be even more thorough experimentation and thus a longer report.

E. Model Size Limitation

You need to be able to store and load your model (weights/parameters) from a file so that it can be evaluated without retraining. The maximum allowed size for your stored model is **50 MB**. Hence, you should try to build a model that is as small as possible while being accurate enough for your dataset.

II. GRADING

The grading of this project is based on two parts: a presentation of your code to your proseminar instructor and the final report.

To complete this project, you have to (1.) provide a version of your model that runs on the JupyterHub to evaluate the *test* dataset, (2.) provide a runnable version of your training scripts as a ZIP file on OLAT (cf. Section IV), (3.) answer questions from your proseminar instructor about your code, and (4.) submit a final report.

On the last proseminar session, your proseminar instructor will spend 5 minutes with each group reviewing the code for your project and ensuring that you have thoroughly understood your assignment. The balance of each partner's contribution will also weigh into this evaluation, so make sure that both partners can thoroughly explain the project.

A final report based on our template must also be submitted. The template contains questions and your report has to provide answers to those questions. We will evaluate your report based on how principled your approach is and how well you are able to answer the questions. As a hint: if you read the report as if it were coming from someone else, would you understand what had been done? Could you reproduce what is described in the report? Does the approach convince you?

III. MISC

- Three years ago we had a leaderboard for bonus points and to compare the methods. This is **not** part of the project this year. However, the code is still in place and in some parts of the code you will see reference to the "leaderboard" test functions. The test data is still accessible through these functions, but now it is transparent to you; before the test data was hidden.
- The dataset is already provided on the hub. **Do not upload any datasets.**

IV. SUBMISSION

There are three different deadlines for this project: one for the coding submission, one for the code presentation, and one for the final report.

For the *code*, **deadline is on June 24th 2026 at 12:00 noon** and you need to submit:

- 1) The runnable training scripts and evaluation scripts as **ZIP** file on **OLAT**
- 2) The working test script on **JupyterHub** as a **notebook submission**

You must present your code on **the June 25th, 2026/June 26th, 2026 during your respective proseminar session**. If

you are unable to attend this proseminar or if you are finished coding earlier, you may request an earlier appointment.

For the *report*, the **deadline is on July 9th 2026 at 23:59** and you need to submit the report as **PDF** file on **OLAT**.

Furthermore, please note the following:

- Your code must be complete and runnable.
- Check that you do not submit a *model that is too big*
- Do not train on the submitted *test*-notebook on JupyterHub! You can create a separate training notebook, or you can use a flag that disables the training (and that is set, so it disables training by default!)
- Do not change the name of the notebook file! Do not replace the notebook! (Because it will be used for testing your model on the test set.)
- We will *deduct 50% of your report points* if your code/notebook submissions not comply with the requirements stated above.